

TABLE I
SNAPSHOT OF THE DEMOGRAPHIC PATTERN OF INDIA

Demographic pattern	Details
Total population	1,210,193,422
Rural population (68.84%)	833,087,662
Urban population (31.16%)	377,105,760
Number of villages	640,867
Number of inhabited villages	597,608
Number of gram panchayats	238,616
Average number of villages per gram panchayat	2.50
Number of blocks*	6612
Average number of panchayats per block	37.43
Number of districts**	640
Total area (in sqkm)	3,287,469

*Block - A group of gram panchayats and associated villages; figure is for year 2012

**District - Geographical division used for administrative purposes

Source: Census of India 2011

the term white space. White space refers to those parts of the licensed spectrum that are not used by the network operator all the time or at all geographical locations.

A block observed to contain generous amounts of white spaces was the TV band, and this was called TV white space (TVWS). TV band spectrum occupies VHF and UHF range and hence has radio propagation properties that make it attractive for communications. VHF/UHF signals travel over longer distances for a given transmission power compared to higher-frequency signals. Thus, large areas can be covered using economical medium-sized antennae.

It has been estimated that to obtain a 100sqkm coverage area using UHF, ten cells are required. This is four times fewer than that needed in 2.4GHz band (43 cells) and 12 times fewer in 5.8GHz band (121 cells). Most countries, including India, are moving from analogue to digital TV transmission, leading to vacating large bands of TV spectrum.

TV band spectrum also has guard bands between channels that, though not contiguous, can be used for wireless communications. The unused TV band spec-

trum may be utilised in the following manner:

Unlicensed. In the same manner as ISM and Wi-Fi bands, but does not address the rural connectivity problem

Auctioning the spectrum. Generates revenue for governments but discourages rural private participation

Unlicensed but controlled access to spectrum. Needs additional technologies to minimise interference to licensed users

In order to guarantee quality of service and spectrum availability to licensed primary users of the spectrum, unlicensed users who first have to identify and occupy unused frequency bands also have to vacate as soon as primary users return. The domain of cognitive radio tackles this through channel sensing and, of late, through geo-location databases that store real-time information about channel availability and usage.

Cognitive radio refers to a device that is aware of its radio environment such as spectrum availability, signal strength, channel access methods, transmission protocols and spectrum policy, and is capable of opportunistically choosing the right parameters to



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